

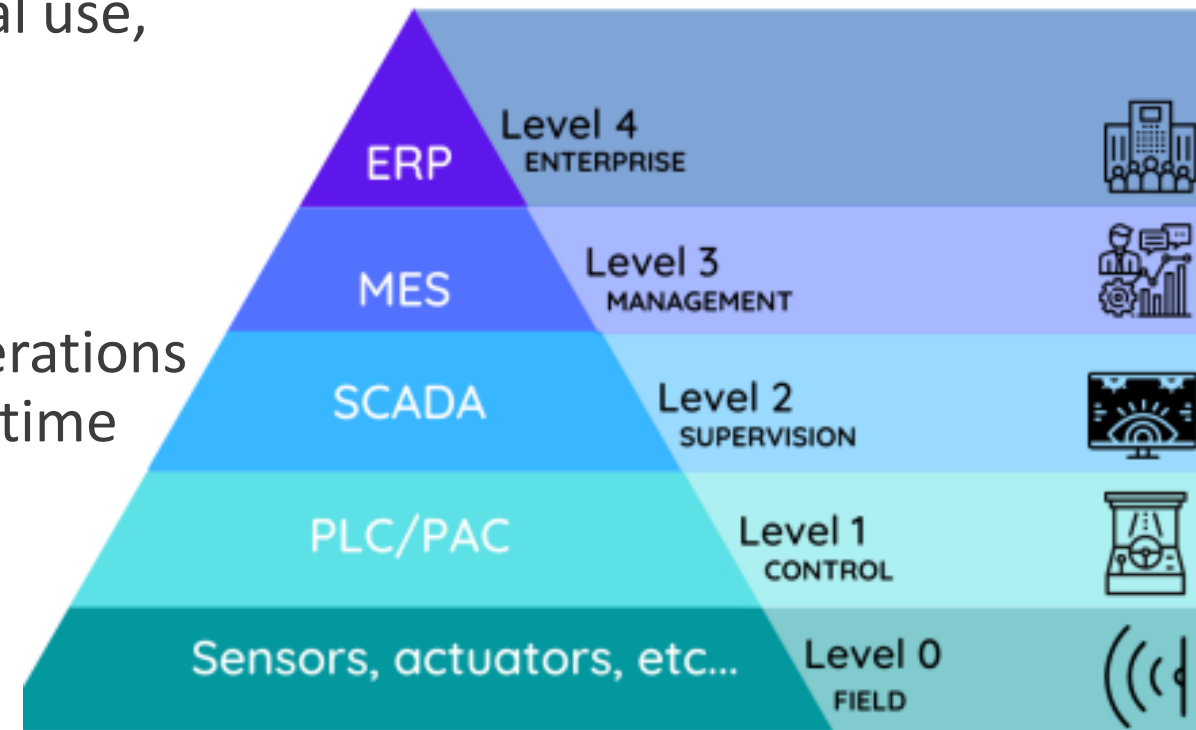
Lecture 9

IoT communication and TwinCAT implementations

- Communication of variables based on name, structures, and objects
 - OPC (classic)
 - OPC-AU
 - REST
 - MQTT
 - CoAP
- IoT Platforms (Middlewares)
- ADS (Beckhoff Protocol that has a lot of support from other vendors)
- TC3 Implementation of ProfiBusDP Master to ABB robot communication
- TC3 Implementation of UPC-UA Server and client

Levels of factory planning and control

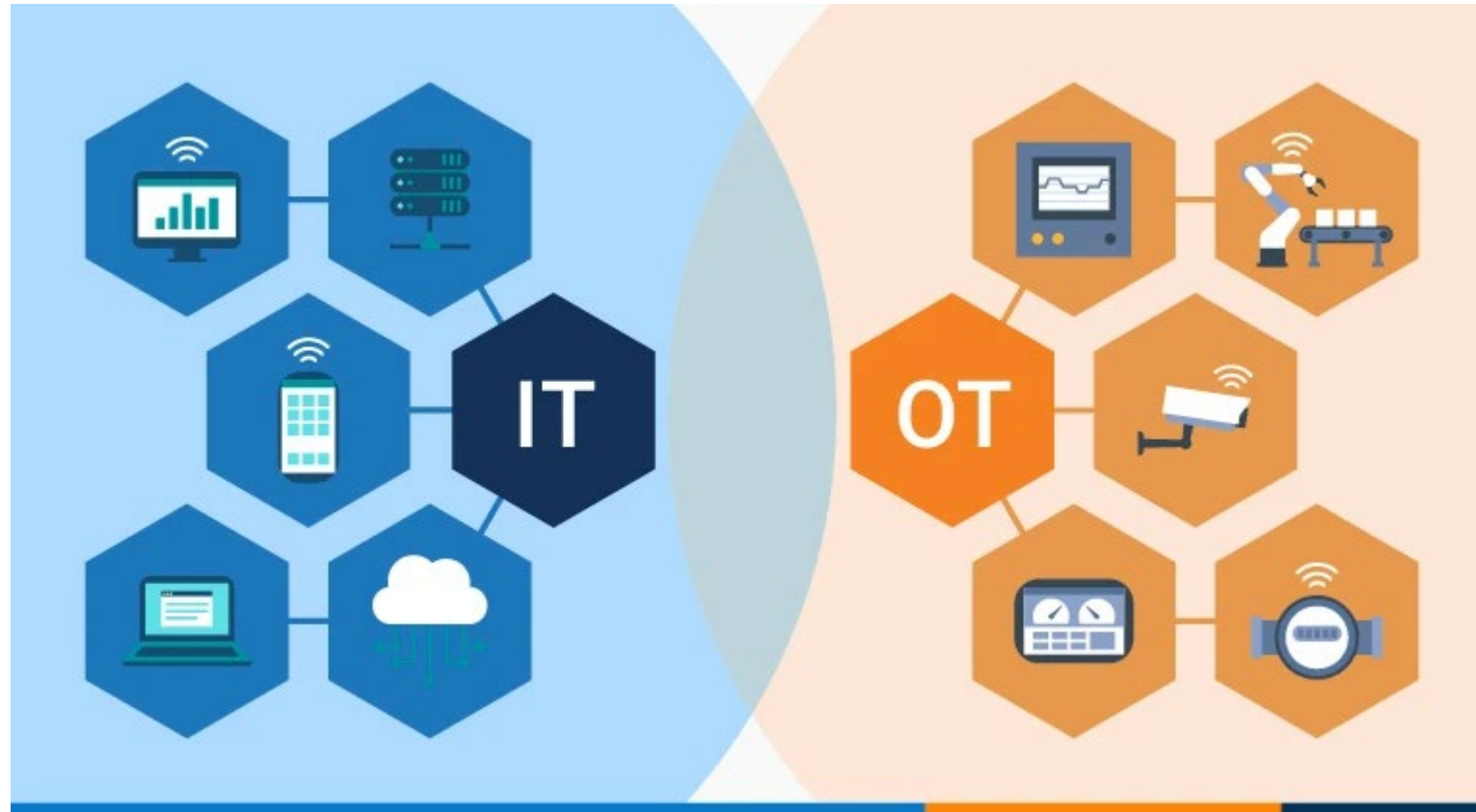
- ERP: Business planning and logistics
 - Plant schedules including production, material use, delivery, and shipping
- MES: Manufacturing Execution System (Middleware)
 - Managing and monitoring manufacturing operations and reporting on production activities in real time
- Process control systems
 - Batch control (SCADA)
 - Control level (PLC, Robot controller, CNC)
 - Field level (Sensors, Actuators, Operators)



Informational Technology (IT) V.S. Operational Technology (OT)

IT: most commonly refers to computers, other endpoint devices like smartphones, servers, and IoT devices.

Servers, networking devices, and endpoint devices



OT: hardware and software systems that directly monitor and control physical processes in industrial environments

Devices typically include industrial computing equipment such as IoT gateways and control systems.

<https://www.onlogic.com/blog/it-vs-ot-how-information-technology-and-operational-technology-differ/>

Data object communication

- Traditional field busses transfer variables such as Boolean, Word, ...
 - Absolute addresses in Real Time (RT)
- Communication protocol that transfers data objects
 - For the industry
 - OLE for Process Control (OPC)
 - Open Platform Communications Unified Architecture (OPC-UA)
 - Internet of Things (IoT) and Industrial Internet of Things (IIoT)
 - Representational State Transfer (REST) (Based on HTTPS)
 - Message Queuing Telemetry Transport (MQTT) (Pub-Sub)
 - Constrained Application Protocol (CoAP)

OPC (classic)

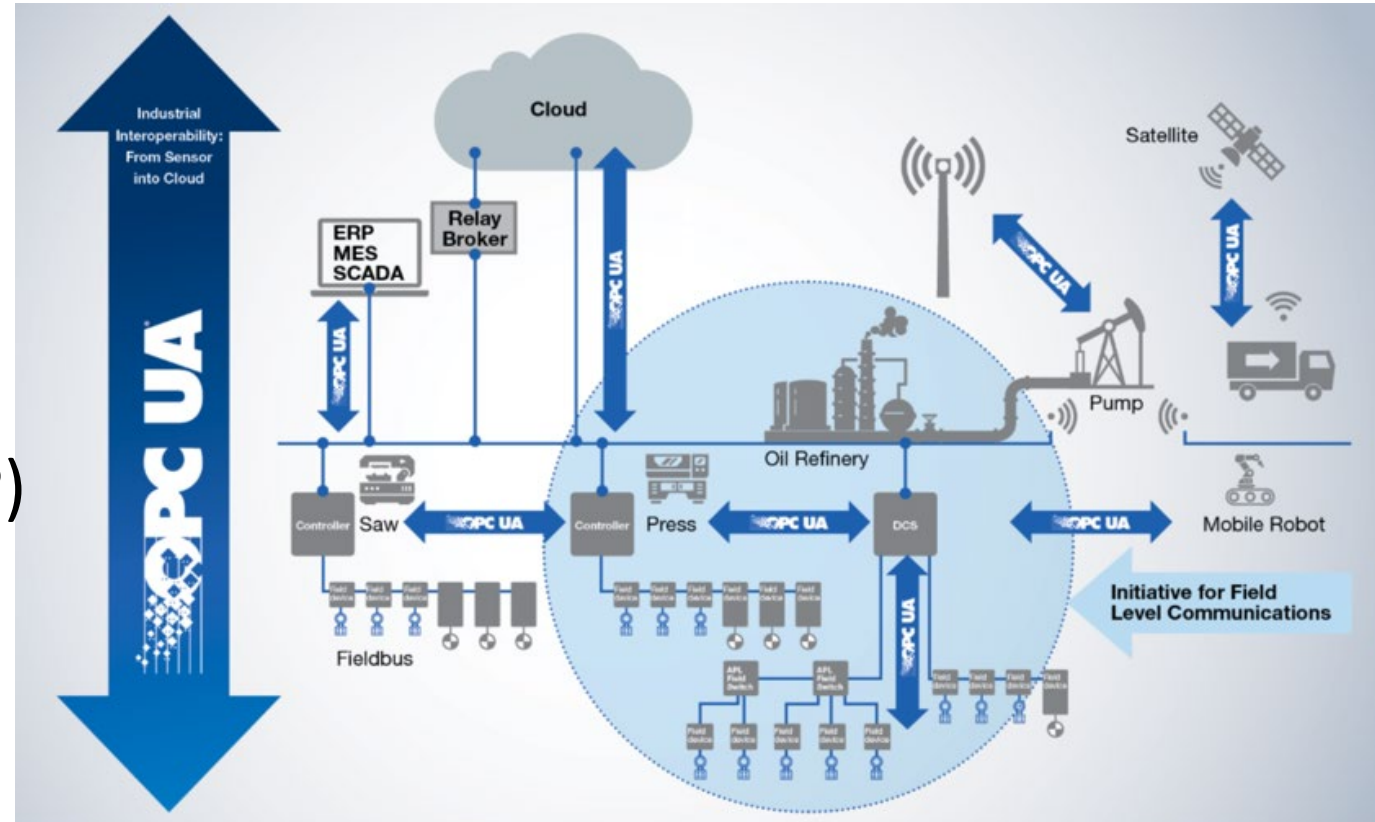
- OLE for Process Control (**Microsoft Windows** implementation)
- OLE: Object Linking and Embedding (Microsoft)
 - since 2011: Open Platform Communications
- An open "standard" for connectivity in industrial automation
 - first version 1996
- OPC Foundation maintains and develops the standards
- OPC, a series of standards for different applications

OPC for different applications

- **OPC Data Access (OPC-DA)**
 - the original and most used
 - data transferring
- OPC Alarms & Events
- OPC Batch
- OPC Data eXchange
- OPC Historical Data Access
 - provides access to data stored with time tags
- OPC Security
 - sensitive information
- OPC XML-DA
- OPC Complex Data
- OPC Commands
- OPC Unified Architecture
 - XML and web services

OPC-UA

- Open Platform Communications Unified Architecture (OPC-UA)
- Communication from sensors to Enterprise Resource Planning (ERP) Manufacturing Execution Systems (MES)
- OPC-UA support
 - Server and Clients applications
 - Publisher and Subscribe (Pub/Sub)



Scalable solutions

Representational State Transfer (REST)

- Web services interface, Client-server architecture
- Object as HTML, XML, JSON
- ABB robots have a REST web service interface

Semantics of HTTP methods	
HTTP method	Description
GET	Get a representation of the target resource's state.
POST	Let the target resource process the representation enclosed in the request.
PUT	Create or replace the state of the target resource with the state defined by the representation enclosed in the request.
PATCH	Partially update resource's state.
DELETE	Delete the target resource's state.
OPTIONS	Advertising the available methods.

Message Queuing Telemetry Transport (MQTT)

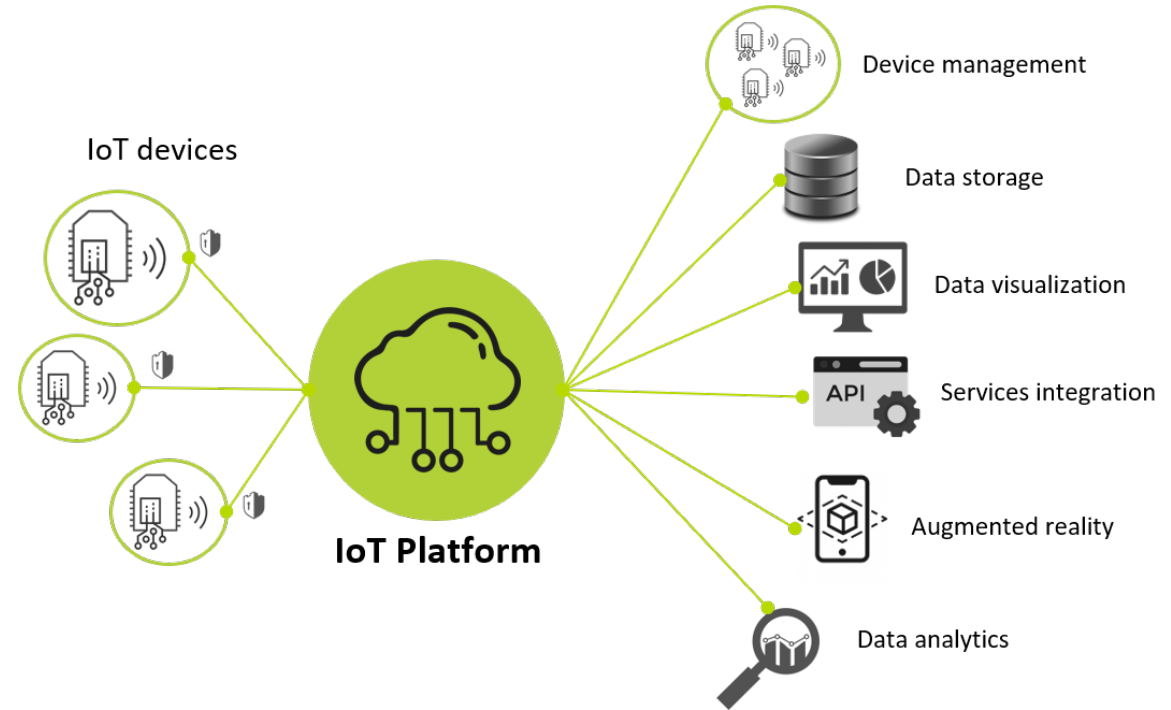
- lightweight machine-to-machine (M2M) protocol
- publish/subscribe model and broker
 - should provide good real-time capabilities
- runs on top of the Transmission Control Protocol (TCP) layer therefore it relies on SSL/TLS for security
- TF6105 | TwinCAT 3 OPC UA Pub/Sub
 - Description: protocol implementation for OPC UA Pub/Sub (UDP and MQTT)
 - <https://www.beckhoff.com/en-en/products/automation/twincat/tfxxxx-twincat-3-functions/tf6xxx-connectivity/tf6105.html>

Constrained Application Protocol (CoAP)

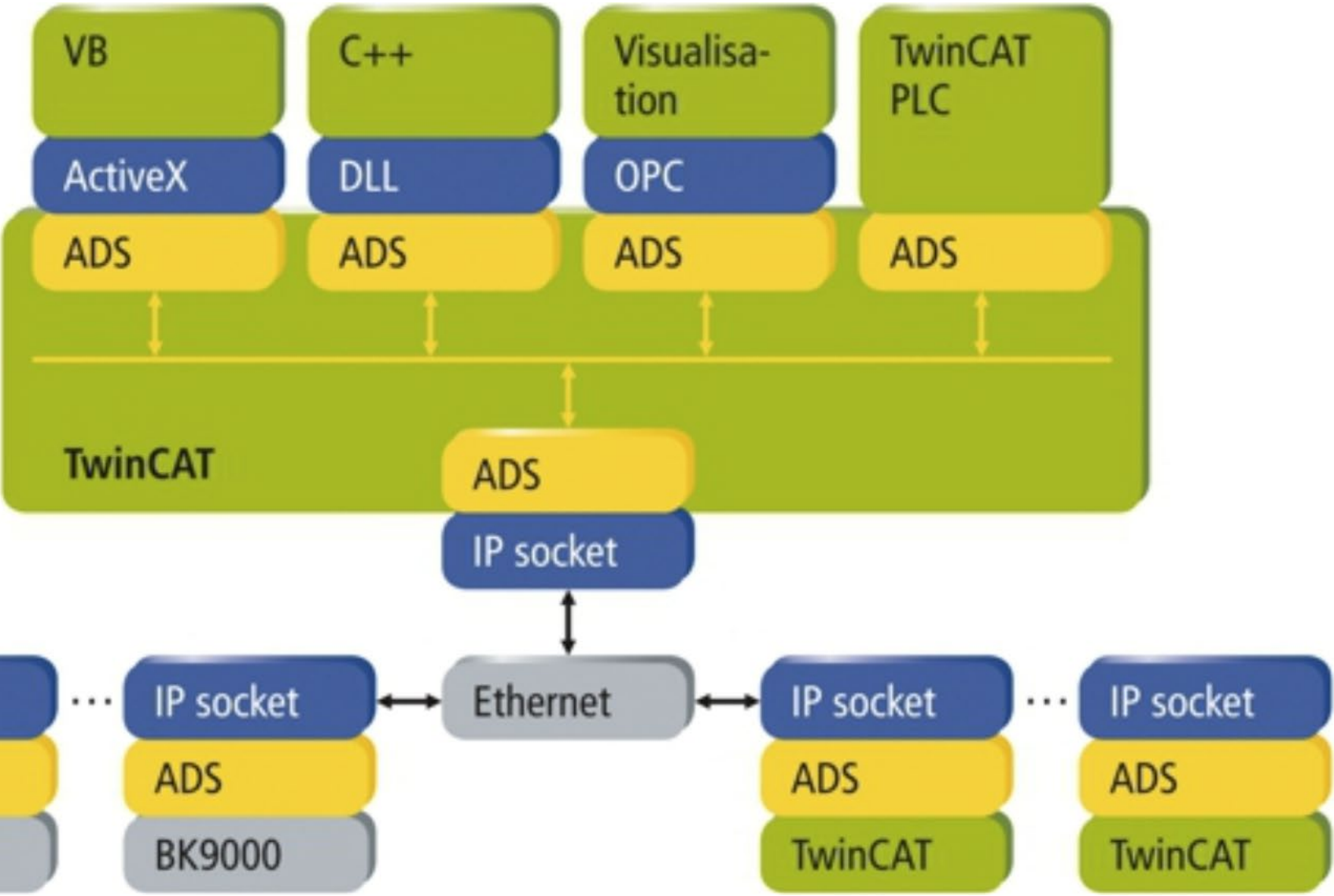
- small low computer power
- based on REST (GET, POST, PUT, and DELETE)
- supports both request/response and publish/subscribe architecture models
- User Datagram Protocol (UDP) layer implies lower overhead

IoT platforms

- Kaa
- ThingsBoards
- ThingSpeak
- ThingWorks
- Nod-RED



ADS
Automation
Device
Specification



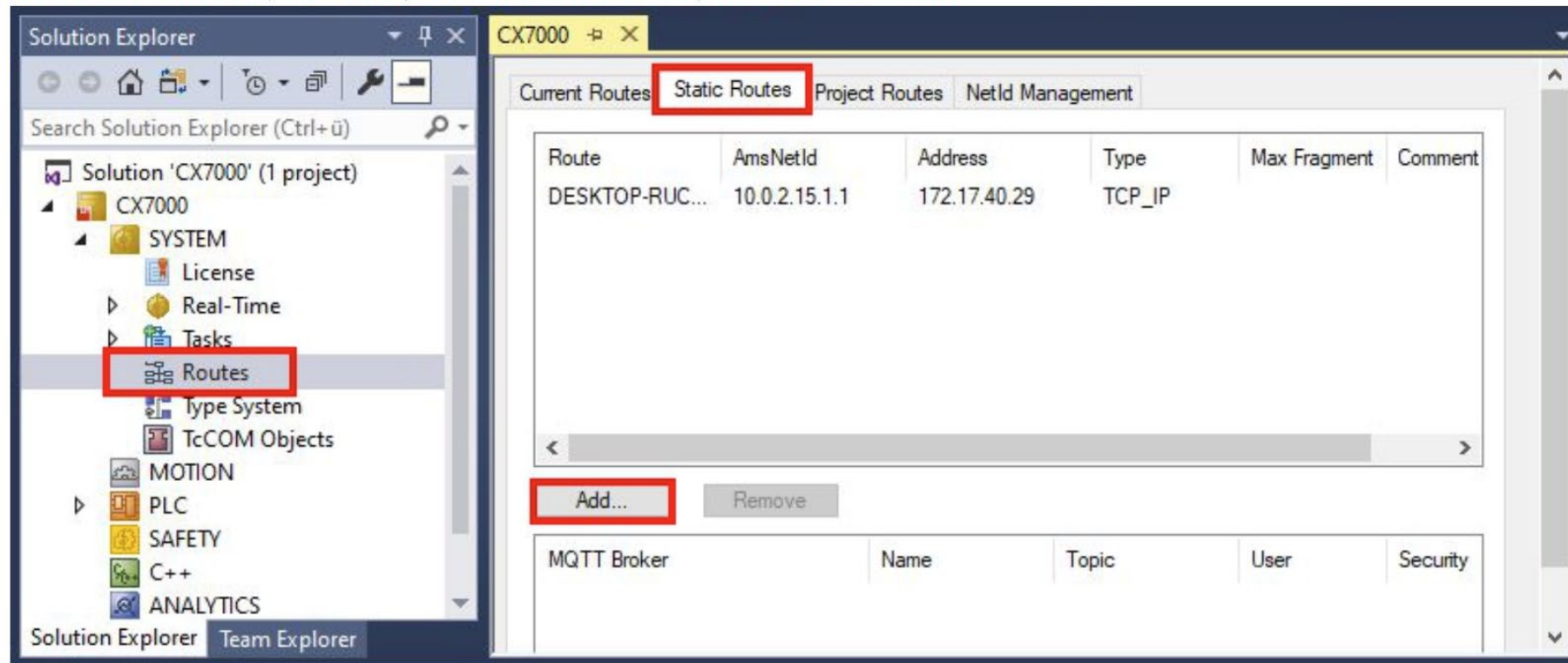
How to set up ADSREAD communication

- [Beckhoff Information System – English](#)
- https://infosys.beckhoff.com/english.php?content=../content/1033/tcplclib_tc2_system/30866571.html&id=

Connect the two systems and add route

Proceed as follows:

1. Start TwinCAT and connect to the first CX70x0 (see: [Connect to the CX70x0](#)).
2. On the left in the tree view, click **Routes**, select the **Static Routes** tab, and click the **Add** button.



3. Under **Remote Route**, select the **Static** option so that the ADS route remains in the project, and then click the **Broadcast Search** button.

Add Route Dialog

Enter Host Name / IP: Refresh Status **Broadcast Search**

Host Name	Connected	Address	AMS NetId	TwinCAT	OS Version	Fingerprint
CX-315394		172.17.40.52	5.49.83.148.1.1	3.1.4022	Windows 7	
CX-35D5DA		172.17.42.34	5.53.213.218.1.1	3.1.4024	Windows 10 1809	3C889B1B41
CX-3D3A1B		172.17.42.23	5.61.58.27.1.1	2.11.2301	Win CE (6.0)	
CX-52D1BA		172.17.42.47	5.85.240.128.1.1	3.1.4024	Windows 10 1809	D08A9839D
CX-5F0F80		172.17.42.27	5.95.15.128.1.1	3.1.4025	TC/RTOS (0.9)	5AC4C9ECE
CX-5F8F6A		172.17.42.28	5.95.143.106.1.1	3.1.4025	TC/RTOS (0.9)	983E7D7F36
CX-5F8F6C		172.17.40.64	5.95.143.108.1.1	3.1.4025	TC/RTOS (0.9)	9E63837915
CX7000-TwinSAFE		172.17.42.21	5.75.184.87.1.1	3.1.4025	TC/RTOS (0.9)	8B0EBA6291

Route Name (Target):

AmsNetId:

Virtual AmsNetId (NAT):

Transport Type:

Address Info:

☐ Host Name ☒ IP Address

Connection Timeout (s):

Max Fragment Size (kByte):

Route Name (Remote):

Target Route: ☐ Project ☒ **Static** ☐ Temporary

Remote Route: ☐ None / Server ☒ **Static** ☐ Temporary

☒ Advanced Settings ☐ Unidirectional

Add Route **Close**

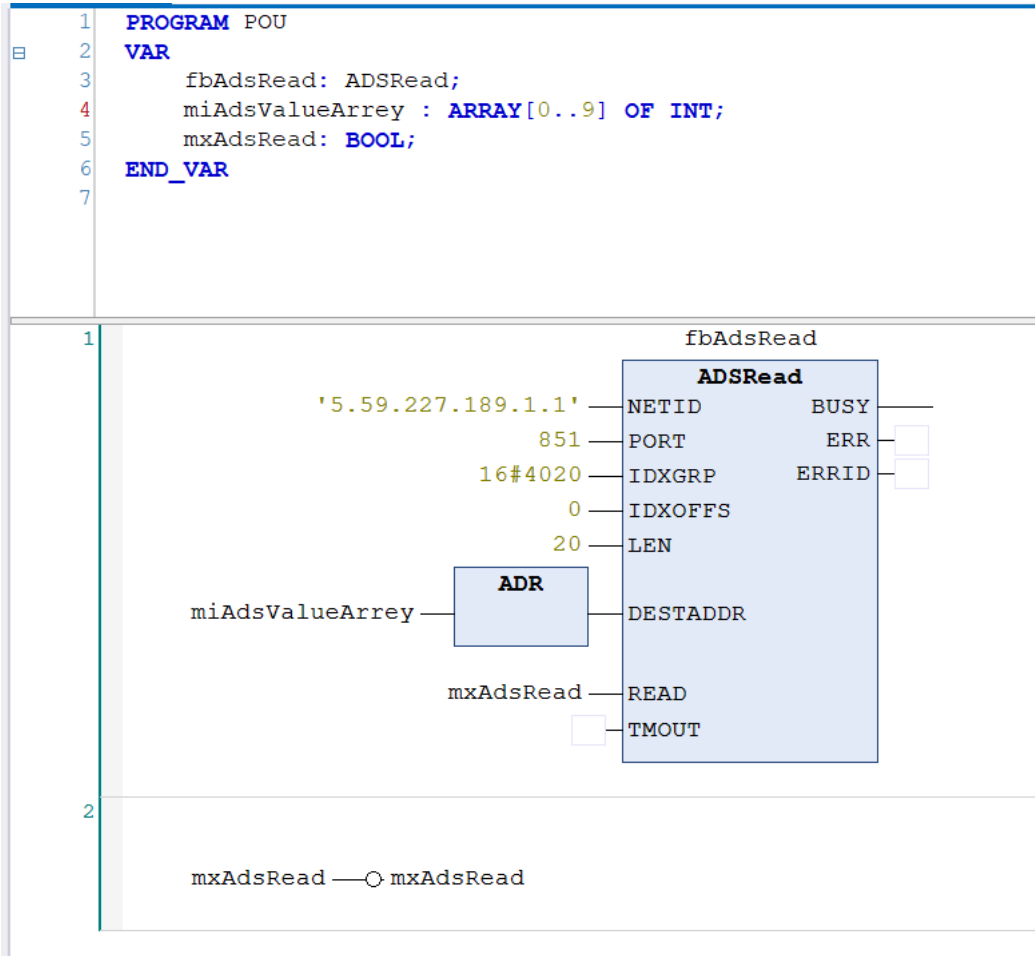
4. Select the second CX70x0 as the destination of the ADS route. The ADS route is entered for both Embedded PCs. The AmsNetId of the second CX70x0 is displayed and can be used in the program for ADS function blocks.

The screenshot displays the Siemens STEP 7 software interface. On the left, the 'Solution Explorer' shows the project structure for 'CX7000', including 'SYSTEM', 'License', 'Real-Time', 'Tasks', 'Routes', 'Type System', 'TcCOM Objects', 'MOTION', 'PLC', 'SAFETY', 'C++', and 'ANALYTICS'. The 'Routes' folder is expanded. On the right, the 'CX7000' window is open, showing the 'Current Routes' tab. The table below lists the current routes:

Route	AmsNetId	Address	Type	Max Fragment	Comment
DESKTOP-RUC...	10.0.2.15.1.1	172.17.40.29	TCP_IP		
CX-5F8F6C	5.95.143.108.1.1	172.17.40.64	TCP_IP		

The 'CX-5F8F6C' route is highlighted with a red box. Below the table, there are 'Add...' and 'Remove' buttons. At the bottom, the 'MQTT Broker' section is visible, with columns for 'Name', 'Topic', 'User', and 'Security'.

PLC 1

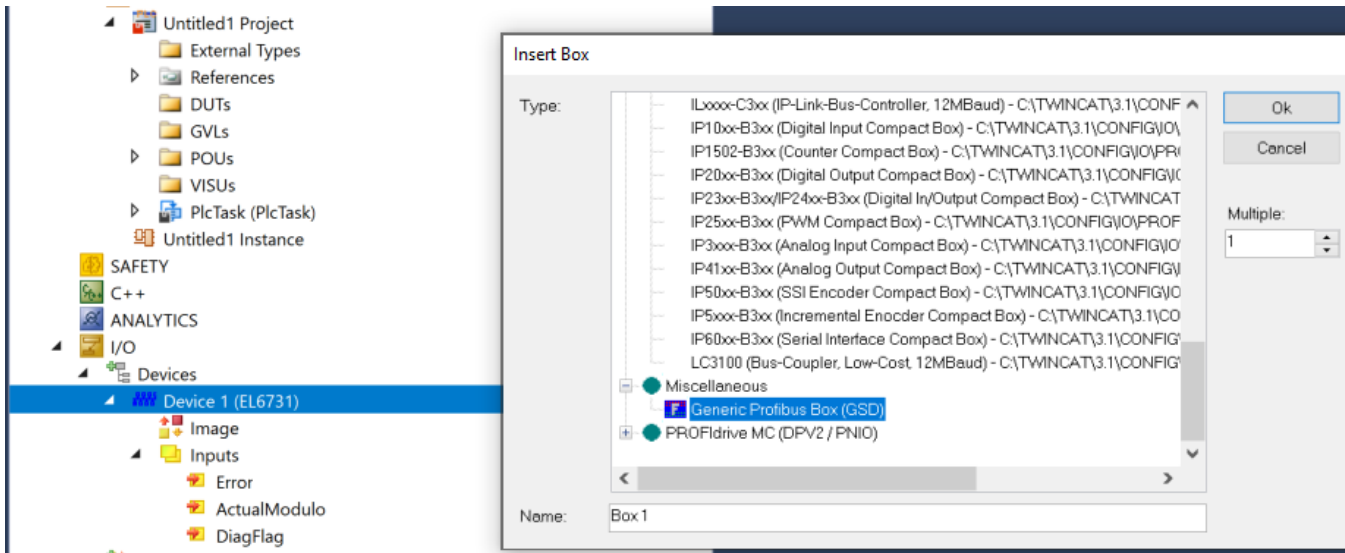


PLC2

```
2 VAR
3
4   miAdsSendValueArray AT %MB0 : ARRAY[0..9] OF INT;
5 END_VAR
6
```

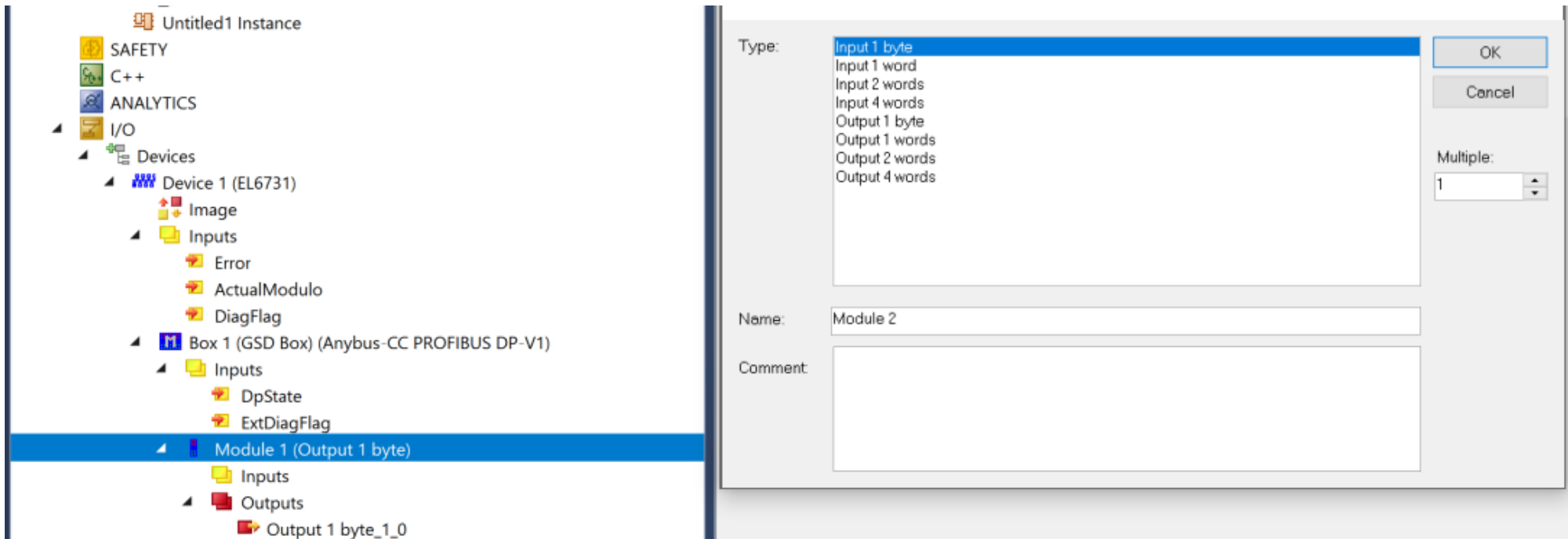
How to add a
Profibus DP Master device
and an ABB robot
Anybus slave device
in
TwinCAT
and In the
ABB robot controller

In TwinCAT, Add Box from GSD file for communication to ABB through anybus adapter



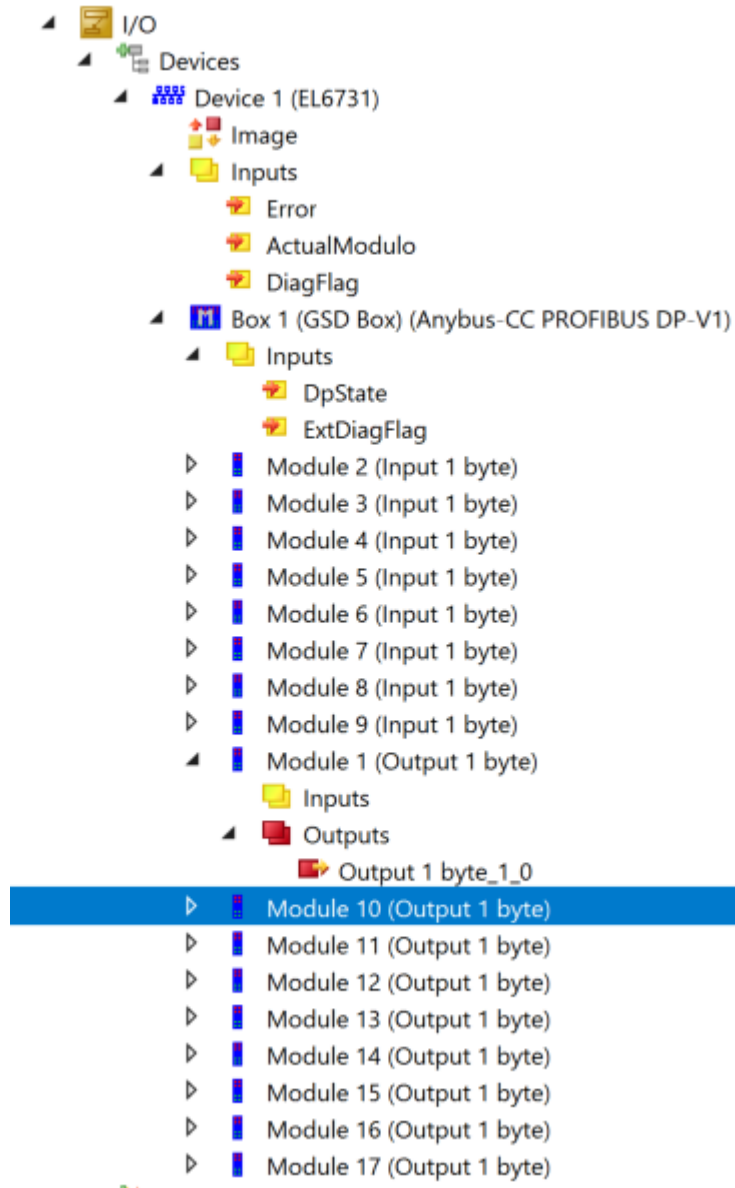
- The Profibus master Device 1 EL6731 will be found by right-clicking on Devices and selecting **scan**, and must be **connected** to the runtime system running in **config mode**
- Right-click on the Profibus device and **add a New Item**
- Select Miscellaneous and **Generic Profibus box (GSD)**, press OK
- Find and select the file **HMS_1811.gsd**, you must have the file on your computer. Click Open

Add input and output bytes and words



- Right-click on Box 1 and Add New Item
- Add words and bytes for inputs and outputs needed according to the setting at the ABB robot
- The PLC and Robot must have **exactly** the **same** settings

8 Input bytes and 8 Output bytes are added



Just an example, your robot may have another setup of I/O (Booleans, bytes, words, double words, ...)

Link the input and output words to variables

The screenshot displays the TwinCAT Project8 MAIN window. On the left, the 'Solution Explorer' shows a project structure with 'I/O' and 'Devices' folders. Under 'Devices', 'Device 1 (EL6731)' is expanded, showing 'Inputs' and 'Outputs'. 'Output 1 byte_1_0' is selected. On the right, the 'Variable' tab shows the configuration for 'Output 1 byte_1_0', which is of type 'USINT', size '1.0', and address '0 (0x0)'. Below this, the 'ADS Info' section shows 'Port: 300, IGrp: 0x8001, IOffs: 0x0, Len: 1'. The 'Full Name' is 'TIID*Device 1 (EL6731)*Box 1 (GSD Box) (Anybus-CC PROFIBUS DP-V1)*Moc'. A dialog box titled 'Attach Variable Output 1 byte_1_0 (Output)' is open, showing a search for variables. The search results list 'MAIN.qbByte1' through 'MAIN.qbByte8', each with a corresponding address and type (e.g., 'QB 385007.0, BYTE [1.0]'). The 'Show Variables' section on the right of the dialog has several checkboxes: 'Only Unused' (checked), 'Exclude disabled' (unchecked), 'Exclude other Devices' (checked), 'Exclude same Image' (checked), 'Show Tooltips' (unchecked), 'Sort by Address' (unchecked), 'Show Variable Groups' (unchecked), and 'Collapse last Level' (checked). The 'Show Variable Types' section has 'Matching Type' (unchecked), 'Matching Size' (checked), 'All Types' (unchecked), and 'Array Mode' (unchecked). The 'Offsets' section has 'Continuous' (unchecked), 'Ignore Gaps' (unchecked), and 'Show Dialog' (unchecked). The 'Variable Name / Comment' section has 'Hand over' (unchecked) and 'Take over' (unchecked). The 'OK' button is highlighted.

The address in the PLC (Station No: 1)

Solution Explorer

- STN200 Instance
 - SAFETY
 - C++
 - I/O
 - Devices
 - Device 1 (EtherCAT)
 - Image
 - Image-Info
 - SyncUnits
 - Inputs
 - Outputs
 - InfoData
 - Term 1 (EK1200)
 - Term 4 (EK1100)
 - Device 2 (EL6731)**
 - Image
 - Inputs
 - Box 10 (GSD Box) (Anybus-CC PROFIB)
 - Module 5 (Input 1 word)
 - Input 1 word_0_0
 - Outputs

TwinCAT Project STN200 | **STN200** | **GVL**

General | **EL6731** | **ADS** | **EtherCAT** | **DP-Diag** | **Box States** | **DPRAM (Online)**

EtherCAT: Term 2 (EL6731) [Search...]

Station No.: 1 [Identify Device...]

Baudrate: 12M [Firmware: no access]

Operation Mode: DP [Firmware Update...]

Cycle Time (µs): 10000 [Hardware Configuration...]

Estimated DP-Cycle (µs): 213 [Upload Configuration]

DP-Cycles/Task Cycle: 1 [Verify Configuration]

☐ Flexible Process Image

☐ Optimized Cycle

☐ Disable Dp-State Modification

Number	Box Name	Address	Type	In Size	Out Size
1	Box 10 (GSD Box) (Anyb...	2	GSD Box	2.0	2.0

The address in the Robot (Station No: 2)

The screenshot displays the TwinCAT software interface. On the left, the 'Solution Explorer' shows a project tree with the following structure:

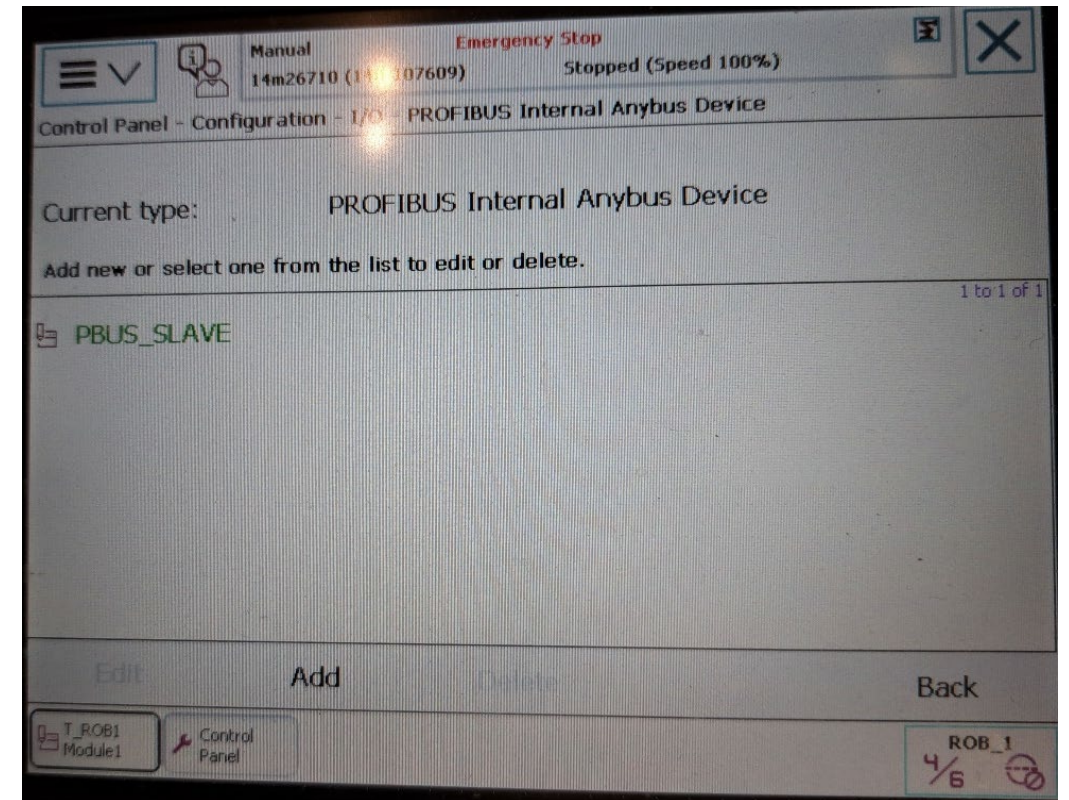
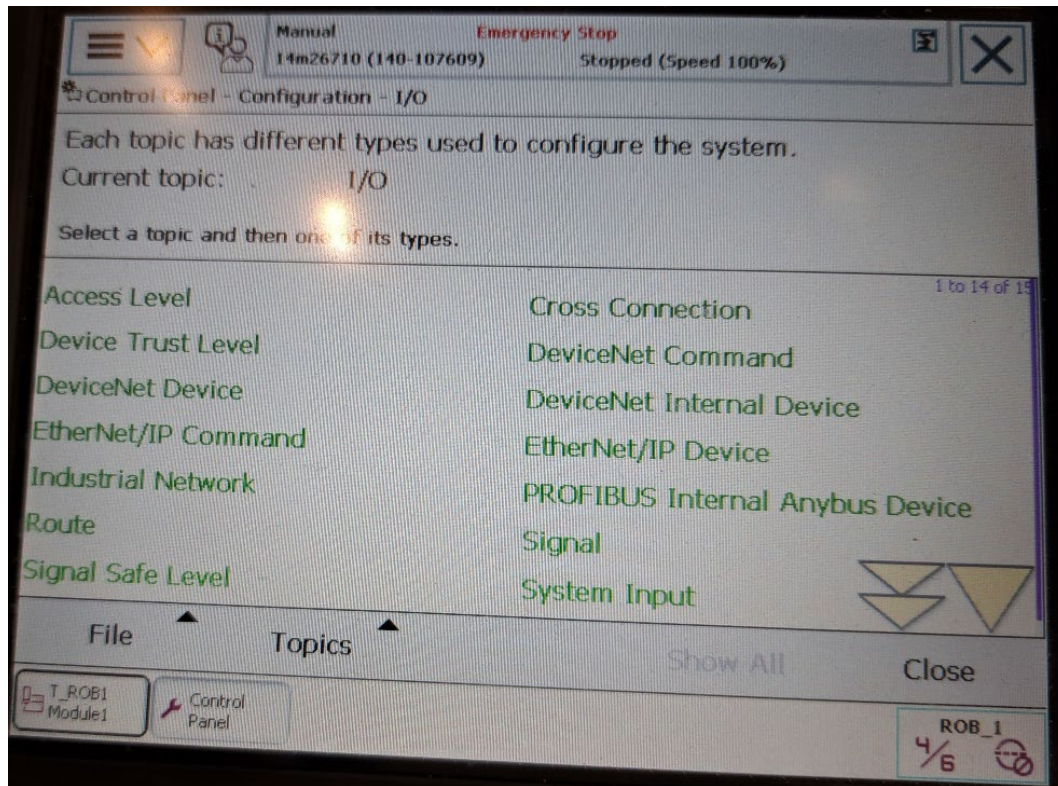
- STN200 Instance
 - SAFETY
 - C++
 - I/O
 - Devices
 - Device 1 (EtherCAT)
 - Image
 - Image-Info
 - SyncUnits
 - Inputs
 - Outputs
 - InfoData
 - Term 1 (EK1200)
 - Term 4 (EK1100)
 - Device 2 (EL6731)
 - Image
 - Inputs
 - Box 10 (GSD Box) (Anybus-CC PROFIB)
 - Inputs
 - Module 5 (Input 1 word)
 - Inputs

The right pane shows the 'General' tab for 'STN200' with the following configuration:

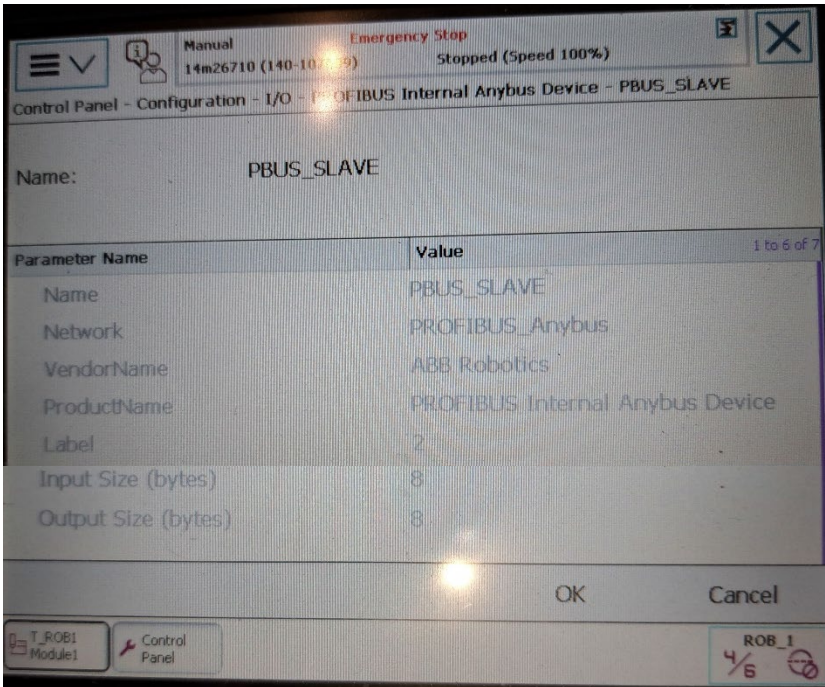
- Station No.: 2 (Set... button)
- Ident No.: 0x1811
- CfgData: D0 E0
- Own PrmData: C0 00 00
- Watchdog:
 - ☒ Enable
 - Time: 200 ms
- DP-/DPV1-Class 1:
 - ☐ Sync/Freeze enable
 - ☐ PrmStruct enable
- Wait for DataExchange:
 - ☐ Enable
 - Timeout (s): 0
- DPV1-Class 2:
 - ☐ Enable DPV1-C2
 - ☐ Disable Abort
 - Watchdog: 1000 ms
- DP-Class 2:
 - ☒ Disabled
 - ☐ Read-Only
 - ☐ No Cyclic Connection
- DPV2:
 - ☐ Publisher enable
 - ☐ Isochron enable
- I/O-Reset Slave button

ABB Robot teach pendant

Control panel, Configuration, and PBUS_SLAVE



Configuration of Profibus AnyBus slave



- The slave addresses “Label” 2
 - Must be the same in TwinCAT
- Input and Output Size in bytes must be the same in TwinCAT.
- OBS! In my slides, it is not the case.
 - TwinCAT has just one word (2 bytes) declared and must have 8 bytes altogether.
 - It can be spread out on bool, bytes, and words but it must be 8 bytes as in the robot

How to set up an OPC-UA Master and Slave in TwinCAT

- Use the Package Manager to download the TF600 package
 - Engineering and Runtime



TF6100 | TwinCAT 3 OPC UA Configurator

graphical configuration tool for TwinCAT OPC UA Server

Engineering



TF6100 | TwinCAT 3 OPC UA Gateway

Allows TwinCAT OPC UA Server aggregation and provides OPC COM DA interface

Runtime



TF6100 | TwinCAT 3 OPC UA Server

OPC UA Server that allows access to symbols from TwinCAT Realtime

Engineering

Runtime



TF610x | TwinCAT 3 OPC UA Client PubSub

Contains TF6100 OPC UA Client and TF6105 OPC UA Pub/Sub

Engineering

Runtime

May not necessarily

TwinCAT OPC UA Server

- supports the following OPC UA functions: Data Access, Historical Data Access, Alarms & Conditions
- Intermediate storage of data on the server: interruption of the communication connection does not lead to loss of data.
- graphical configurator for simple handling of local and remote OPC UA servers
- definition of user/groups-based access rights at namespace and node level
- support of client/server certificates to establish secure communication
- support of the TwinCAT [Analytics](#) file format for Historical Access

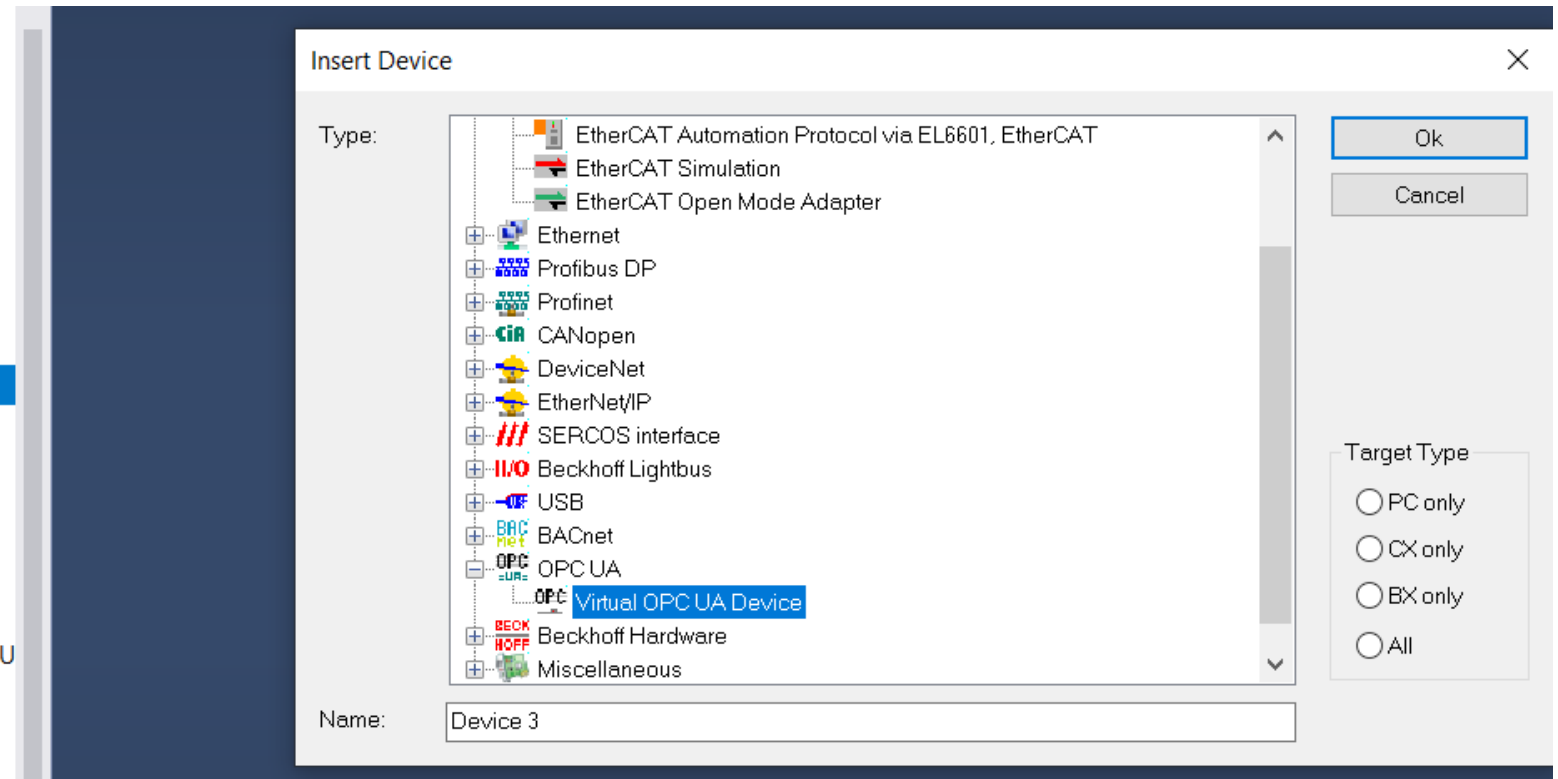
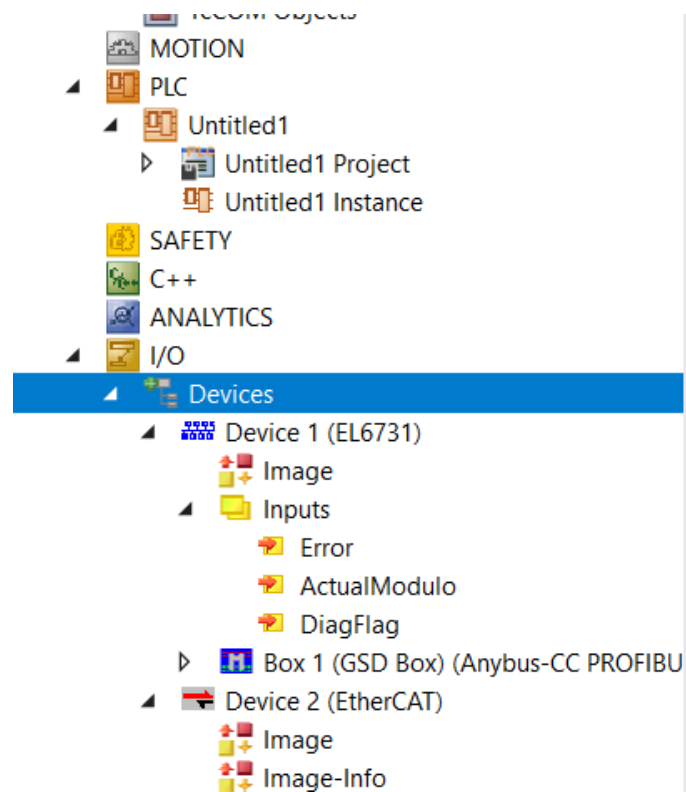
TwinCAT OPC UA Gateway

- high-performance, free-of-charge OPC COM DA server
- enables aggregation of multiple, subordinate TwinCAT OPC UA servers
- support of client/server certificates to establish secure communication

TwinCAT OPC UA Client

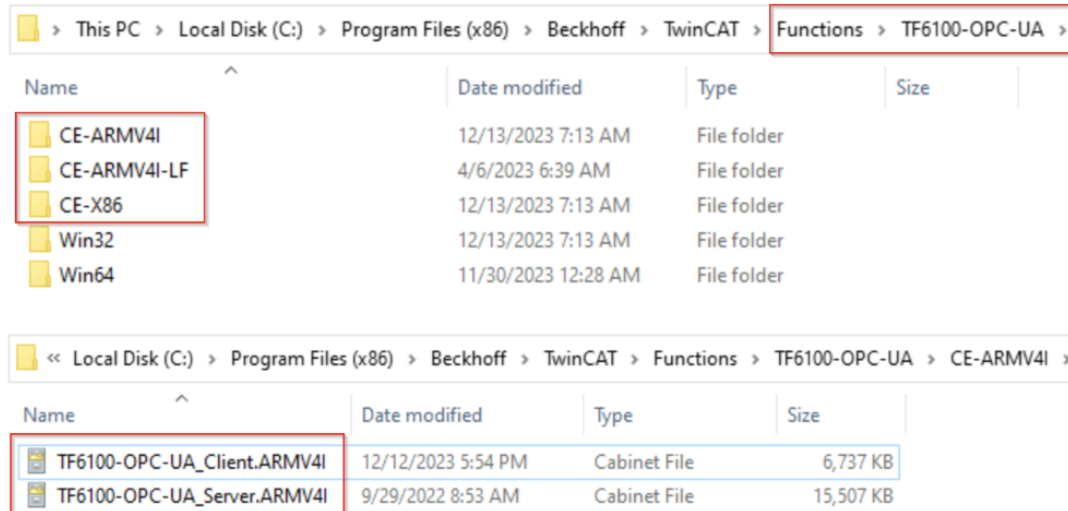
- supports OPC UA communication with remote OPC UA servers
- available as PLCopen-compliant PLC function block or as TwinCAT [I/O](#) device
- support of client/server certificates to establish secure communication
- automatic code generation

Now you can find and add a virtual OPC UA Device



Windows CE

If you are using Microsoft Windows CE as your operating system, you can install this function via the respective CAB files, which are delivered with the setup or TcPkg workload. The CAB files are usually stored in the subdirectory CE-ARMV4I and CE-X86 relative to the installation directory of the function.



From there they can be transferred to the Windows CE device via file transfer and executed there. The CAB files then install and register the function on the respective system.

Always use the appropriate CAB file for your system. Specifically, this means

- CE-ARMV4I: Arm®-based devices, e.g. CX8190, CX9020
- CE-X86: x86-based devices, e.g. CX51xx, CX52xx, CX20xx

The CAB file can be transferred to the device either via the CF/SD card or the FTP server integrated in Windows CE.



Device restart

After installing this function, a device restart is required so that the function can be used.

https://infosys.beckhoff.com/english.php?content=../content/1033/tf6100_tc3_opcu_client/index.html&id=

Declare OPC-UA variables in the server PLC

VAR

```
// OPC-UA Communication to STN500  
{attribute 'OPC.UA.DA' := '1'}  
iToSTN500FromSTN200 : ARRAY[1..8] OF INT;  
{attribute 'OPC.UA.DA' := '1'}  
iFromSTN500ToSTN200 : ARRAY[1..8] OF INT;
```

END_VAR

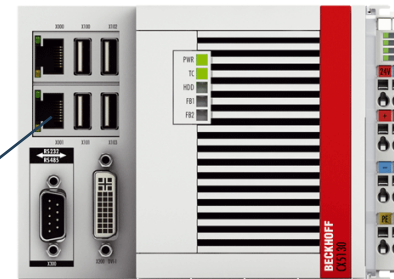
- The **attribute** 'OPC.UA.DA' := '1' makes the variable or object (function block or data structure) **visible** in the OPC UA client PLC

Update OPC-UA in the target system

- The same version of TF6100 for OPC-UA must be installed in the target system as well as on the engineering PC.
- Check the firewall for OPC-UA communication on all involved systems!
- Use the Beckhoff software “Sample Client” to test the communication against servers

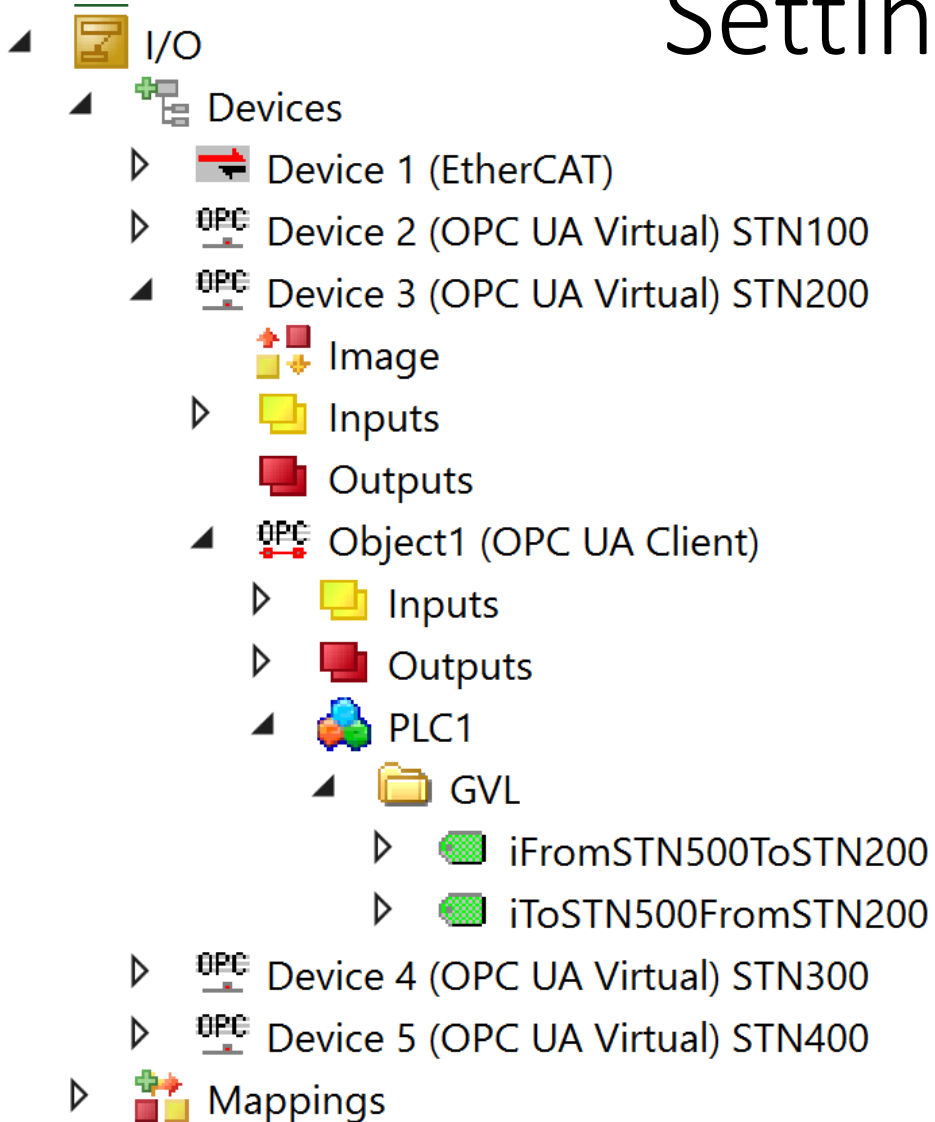


Engineering PC



Target IPC

Setting up an OPC-UA client PLC



- Right-click on “Devices”
- Add an OPC-UA client
- If the systems is connected, it **will find** the OPC-UA server
- Select the PLC1 for PLC variables

Link data to variables

Solution Explorer

Search Solution Explorer (Ctrl+⁺)

Device 2 (OPC UA Virtual) STN100

Image

Inputs

Outputs

Object1 (OPC UA Client)

Inputs

Outputs

PLC1

GVL

iFromSTN500ToSTN100

Inputs

Value

Outputs

Value

Value[0]

Value[1]

Value[2]

Value[3]

Value[4]

Value[5]

Value[6]

Value[7]

iToSTN500FromSTN100

Inputs

Value

Value[0]

Value[1]

Value[2]

Value[3]

Value[4]

Value[5]

Value[6]

Value[7]

TwinCAT Project STN500

STN500

Variable

Flags

Online

Online List

Name: Value

Type: ARRAY [0..7] OF INT

Group: Outputs

Size: 16.0

Address: 2 (0x2)

User ID: 0

Linked to... GVL.iFromSTN500ToSTN100 . PlcTask Outputs . STN500 Instance . STN500

Comment: From STN500

ADS Info:

Full Name: TIID^Device 2 (OPC UA Virtual) STN100^Object1 (OPC UA Client)^PLC1^GVL^